

The neural bases of distracter-resistant working memory

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Abstract A major difference between humans and other animals is our capacity to maintain information in working memory (WM) while performing secondary tasks, which enables sustained, complex cognition. A common assumption is that the lateral prefrontal cortex (PFC) is critical for WM performance in the presence of distracters, but direct evidence is scarce. We assessed the relationship between fMRI activity and WM performance within subjects, with performance matched across distracter and no-distracter conditions. Activity in the ventrolateral PFC during WM encoding and maintenance positively predicted performance in both conditions, whereas activity in the presupplementary motor area (pre-SMA) predicted performance only under distraction. Other parts of the dorsolateral and ventrolateral PFCs predicted performance only in the no-distracter condition. These findings challenge a lateral-PFC-centered view of distracter resistance, and suggest that the lateral PFC supports a type of WM representation that is efficient for dealing with task-irrelevant input but is, nonetheless, easily disrupted by dual-task demands.

Edward E. Smith (deceased August 17, 2012) contributed to this work through his close and invaluable involvement in study design, interpretation of results, and manuscript preparation.

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Working memory (WM) refers to a system that maintains currently relevant goals and information for use in guiding ongoing information processing. Information in WM is thought to guide the deployment of attention (de Fockert, Rees, Frith & Lavie, 2001) and the manipulation of information to achieve current task goals (Baddeley, 1986; Hitch & Baddeley, 1976). Probably no factor has more effect on WM performance than the presence of distracters (e.g., Keppel & Underwood, 1962). Consider the simple task of remembering a group of three letters—a trigram—over an interval of a few seconds. In a classic article (Peterson & Peterson, 1959), participants rehearsed a trigram for intervals of up to 18 s while counting backward by threes. With this distracting secondary counting task, what would otherwise have been trivial to remember became virtually impossible, with near-zero recall by 18 s, with proactive inhibition amassed from prior trials having been shown to underlie this type of memory impairment (Keppel & Underwood, 1962). The powerful effect of distraction is evident in everyday situations: Memory for the contents of a book or a phone number just memorized may be obliterated by a few moments of idle conversation.

Distracters in WM are of two basic types. They may not require any intentional processing, in which case they are simply perceptual experiences to be ignored while remembering other items, or they may constitute a secondary task that must be performed during maintenance of the memory set. Successful inhibition of merely perceptual distracters may rely on perceptual filtering mechanisms that can operate at an early processing stage (Bundesen, 1990; Treisman, 1985), whereas dual-task interference is likely to rely more heavily on task switching, coordination, and information selection processes. Performance on tasks involving these two distracter types is essentially uncorrelated (Miyake, Friedman, Emerson, Witzki,

Howerter & Wager, 2000). Crucially, performance on WM tasks with dual-task distracters, rather than perceptual distracters, predicts performance on complex tests of fluid intelligence (Conway, Cowan, Bunting, Theriault & Minkoff, 2002; Conway, Kane & Engle, 2003; Engle, Kane & Tuholski, 1999; Fry & Hale, 1996; Kane & Engle, 2002, 2003), correlates with other indices of “executive function” (Bleckley, Durso, Crutchfield, Engle & Khanna, 2003; Kane & Engle, 2003), and predicts everyday cognitive failures (Friedman & Miyake, 2004).

Although there have been many studies investigating the brain mechanisms of selective attention and perceptual distracter suppression, very few studies have investigated those involved in creating WM representations resistant to dual-task distracters. The prefrontal cortex (PFC) is thought to be crucial for distracter resistance (Baddeley & Della Sala, 1996; Knight, Richard Staines, Swick & Chao, 1999; Miller & Cohen, 2001), but much of the key evidence has come from studies of perceptual distracters (Chao & Knight, 1995, 1998; Jha, Fabian & Aguirre, 2004; Malmo, 1942; McNab & Klingberg, 2008; Miller, Erickson & Desimone, 1996; Postle, 2005; Sreenivasan & Jha, 2007; Tsushima, Sasaki & Watanabe, 2006). In an important study, Sakai, Rowe and Passingham (2002) examined the basis of resistance to dual-task distraction by assessing the relationship between PFC activity during WM encoding and maintenance with recall accuracy after performing a distracting secondary task. Activity during encoding/maintenance in right Brodmann’s area (BA) 46, which spans the dorsolateral and ventrolateral prefrontal cortices (DLPFC/VLPFC), was greater for correct than for error trials. On the basis of this finding, suggested that this area is critical for distracter-resistant memory.

However, the Sakai et al. (2002) study was limited in a way that substantially undermines its conclusion. Although they showed an accuracy effect in the presence of a distracting task, they did not demonstrate that the accuracy effect was *specific* to the distraction condition. They did include a no-distraction condition, but participants made virtually no errors without distraction. Thus, the alternative remains that the lateral PFC plays a general role in WM maintenance with or without distraction. In addition, the accuracy probe on the WM task involved a binary “yes or no” recognition decision, precluding the possibility of assessing parametric relationships between brain activity and accuracy, which would help establish that there is a meaningful relationship between PFC activity and accuracy across the normal range of performance.

Two plausible alternatives for the role of the lateral PFC provide the basis for hypotheses in the present study. First, the lateral PFC may be important for WM maintenance with or without distracters. In several studies (D’Esposito & Postle, 1999; Ptito, Crane, Leonard, Amsel & Caramanos, 1995), patients with lesions of BA 46 were impaired on delay tasks with or without distraction. Similarly, monkeys with

prefrontal lesions have shown impaired performance on WM tasks with and without distracters (Funahashi, Bruce & Goldman-Rakic, 1989; Funahashi, Inoue & Kubota, 1997; Sawaguchi & Yamane, 1999), although it has been argued that monkeys are distractible even without explicit distracters included in the task. Second, the lateral PFC may be particularly important for filtering out and reducing interference from perceptual distracters—as evidenced by neuroimaging studies (de Fockert et al., 2001; Dolcos, Miller, Kragel, Jha & McCarthy, 2007; Jha et al., 2004; Postle, 2005; Sreenivasan & Jha, 2007), humans and monkeys with prefrontal damage (Chao & Knight, 1995, 1998; Malmo, 1942), and primate electrophysiology (Artchakov, Tikhonravov, Ma, Neuvonen, Linnankoski & Carlson, 2009; Miller et al., 1996; Tsushima et al., 2006)—but this region may not be crucial for resistance to dual-task distraction. This idea is consistent with evidence for a broad role for the VLPFC in particular in the selection of task-relevant sensory stimuli (Badre, Poldrack, Pare-Blagoev, Insler & Wagner, 2005; Fuster, 2008; Nee & Jonides, 2009; Thompson-Schill et al., 2002). Thus, resisting perceptual distracters may load heavily on stimulus selection, whereas resisting dual-task distraction may load heavily on coordination and scheduling of WM processes and, thus, place demands on different cortical areas.

We studied resistance to dual-task distraction using a standard verbal WM task with word memoranda and several novel features (Fig. 1). First, like Sakai et al. (2002), we included both distracter and no-distracter conditions. In the distracter condition, there was a delay before onset of the secondary distracter task so that we could assess brain activity *before* distraction and relate it to subsequent accuracy. Second, to provide a parametric assessment of accuracy on each trial, we included a series of four yes/no recognition probes after each trial, resulting in five levels of accuracy (0–4 correct). Finally, to permit a sensitive analysis of the brain–accuracy relationship for *both* distracter and no-distracter conditions, we used an adaptive staircase procedure to select a memory set size for each subject and each condition that resulted in approximately 75 % accuracy (3/4 probes correct on average) in both conditions. This allowed us to compare brain–accuracy relationships for distracter versus no-distracter conditions, testing whether the lateral PFC and/or other areas show monotonic increases in activity with performance only under dual-task distracter conditions.

Method

Participants

Seventeen healthy right-handed participants (mean age = 22.3 years, 6 female) were recruited in compliance with the human subjects regulations of Columbia University and

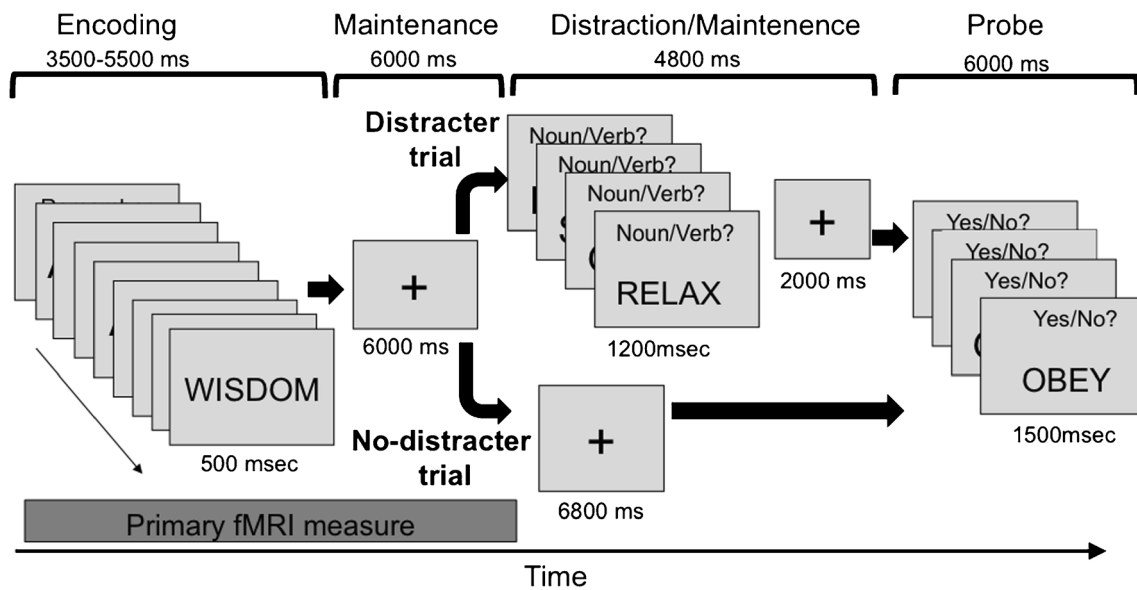


Fig. 1 Task design. During encoding, memoranda were presented every 500 ms, with the number of words for each participant determined before the main task using an adaptive procedure targeting 75 % accuracy. During maintenance, participants rehearsed the memoranda. On distracter trials, maintenance was followed with a secondary task in which

participants made noun/verb judgments on four words. On no-distracter trials, participants continued maintaining the items for the same duration as the secondary task on distracter trials (6,800 ms). In the probe period, participants saw four words and made yes/no judgments as to whether they were part of the memory set

provided informed consent in accordance with the Declaration of Helsinki. They were paid \$20/h for voluntary participation, plus bonuses as described below. Handedness was assessed with the Edinburgh Handedness Inventory, and eligibility was assessed with a general health questionnaire and fMRI safety screening form.

Materials and procedures

Stimuli

Stimuli were selected from a list of two-syllable words ranging from four to six letters in length (mean 5.5), generated using the online version of the MRC Psycholinguistic Database (http://www.psy.uwa.edu.au/mrcdatabase/uwa_mrc.htm). Fifty nouns and 50 verbs were then each selected randomly from that list for use in the experiment. The words had intermediate average levels of concreteness (mean=335), imageability (mean=357), and Kučera–Francis frequency (mean=85). Words included in each trial were selected randomly with replacement from the list of 100 words. Items from this list could appear in both the encoding and the distraction events of the trial, with the constraint that items in the encoding event could not appear in the distraction event of the same trial. Due to the randomization procedure, items appeared approximately equally in the task, and a single trial included, on average, 50 % nouns and 50 % verbs. There were no restrictions on the number of nouns/verbs appearing in the encoding or distraction events.

Experimental structure

The experiment consisted of 80 WM trials. As is shown in Fig. 1, on each trial, participants intentionally encoded a series of words (“memory set”). The set size was chosen for each individual on the basis of a prescan calibration (described below). During the encoding phase, each word was presented on a computer screen for 500 ms (with an approximate visual angle of 1°). Participants then maintained the word set over a 6,000-ms maintenance period, during which a fixation cross was presented on the computer screen. How fMRI activity varied during these combined encoding and maintenance periods was of primary interest.

Following maintenance, trials were divided into two types, which occurred with equal frequency. On distraction (D) trials, participants performed a secondary judgment task. They viewed a series of four words, each presented for 1,200 ms, and indicated whether each distracter was a noun or a verb by pressing the index or middle finger of the right hand, respectively, on an MRI-compatible button box. No words were members of the memory set. Participants were instructed to maintain the original memory set during the secondary task. A 2,000-ms fixation period followed the set of distracters. On no-distraction (ND) trials, participants viewed a fixation cross for the same amount of total time as the distraction task (6,800 ms) and continued to maintain the memory set.

D and ND trials were grouped in blocks of 10 trials, so that an entire scanning run consisted of either D or ND trials. Participants were told in advance that the trials would be

blocked and, therefore, had information that would allow them to prepare differently during encoding/maintenance for D versus ND trials.

Following the continued maintenance with or without the distracter task, participants viewed a series of four unique probe words, each presented for 1,500 ms. Participants indicated whether each word was part of the memory set with a “yes” (index finger) or “no” (middle finger) response on the button box. This multistage probe period allowed for five levels of accuracy (0–4 probes correct) on each trial, providing a more sensitive measure of WM performance than has previously been used. Probes were randomly selected from the memory set and from a set of words that was not part of the memory set (two of each). Items were selected at random and were, therefore, on average approximately equally sampled. Sampling of items from the memory set was not stratified on the basis of serial position of presentation. Although a serial rehearsal strategy could have been used by participants (since the items were likely rehearsed in some order, and using presentation order would be plausible), it would not confer a particular advantage at test. Following the probe period, a 20 s intertrial interval allowed fMRI responses to return to approximate baseline values, which permitted us to quantify activity on individual trials.

Performance incentives

Participants were incentivized with instructions that they could receive a \$0.10 monetary reward on every trial for successful performance, with the potential to earn an additional \$8.00 over the course of the experiment. On ND trials “successful” performance for reinforcement purposes was defined solely in terms of memory performance, whereas D trials also required three or four correct responses to the four noun/verb or yes/no judgments. Also, on D trials, participants received equivalent monetary incentives (\$0.05 each) for good performance on both the probe and the distracting noun/verb judgments to ensure that they would attend to both tasks. Participants did not receive any feedback about monetary earnings until the end of each block (scanning run). Bonuses ranged from \$3.70 to \$5.30, with an average of \$4.61.

Memory set titration

D and ND trials were matched for difficulty within and across participants during a titration phase that occurred before fMRI scanning. This titration phase also served to familiarize the participants with the 100 words used in the task to eliminate novelty effects. Participants performed the task using memory sets of varying sizes until they were achieving successful performance on 75 % of the trials. The criterion of 75 % correct (three of four probes) was chosen because it lies midway between chance performance (50 %) and perfect

accuracy (100 %) and, thus, maximizes power to detect accuracy effects by avoiding either ceiling or floor effects. Statistical power is maximized when there are equal numbers of trials at both ends of the useful range of an outcome (here, 2/4 trials [chance] vs. 4/4 trials [perfect]), since the predictor variance is maximal.

The titration task was 40 trials long, with one block of D and ND trials each (20 trials per block). The task started with eight words in the memory set of the D condition and nine words in the memory set of the ND condition. After every 5 trials, set size updating occurred such that the size increased by one if accuracy exceeded 75 % and decreased by one if accuracy fell below 75 %. After all trials, the final set size was determined by regressing trial-by-trial accuracy on trial-by-trial set size and then by using the regression line to identify the size associated with 75 % correct.

Across participants and conditions, memory set size varied from 7 to 11 words. For each of the 17 participants, the titrated memory set size was larger in the ND condition (mean=10.77 words, $SD=0.56$) than in the D condition (mean=9.24 words, $SD=0.75$), indicating that the distracting noun/verb judgment task was having a deleterious effect on WM ability (mean difference=1.53 words), $t(16)=12.25$, $p<.001$. These set sizes were used in the subsequent fMRI task. Thus, the comparison in brain activity for [ND–D] trials differed in the number of words encoded, and might be expected to produce an overall [ND>D] difference in standard WM networks (although if encoding/maintenance processes were more strongly engaged during D blocks, it would tend to offset the set size effect by creating a [D>ND] difference). However, the [ND–D] comparison was not of primary interest; we designed the study to focus on relationships between brain activity and accuracy within each condition. The titration ensured a homogeneous distribution of accuracy values across participants and a distribution across a range of accuracy values within participants, for both the D and ND conditions, which is desirable for sensitive detection of brain–accuracy relationships.

Data acquisition

Whole-brain fMRI data were acquired on a 1.5 T GE Signa Twin Speed Excite HD scanner (GE Medical Systems). Functional and anatomical images were acquired with a T2*-sensitive EPI BOLD sequence with a TR of 2,000 ms, TE of 41 ms, flip angle of 60°, ascending interleaved acquisition, field of view of 22 cm, 24 slices, and $3.44 \times 3.44 \times 4.5$ mm voxels, with whole-brain coverage. Stimulus presentation and data acquisition were controlled using MATLAB (<http://www.mathworks.com/>) and the Psychophysics Toolbox (<http://psychtoolbox.org/>). An LCD projector displayed stimuli on a back-projection screen mounted in the scanner suite. Responses were made with the right hand via a

five-finger button response unit with a molded hand brace (Resonance Technologies, Inc.).

Image processing and data analysis

Image denoising

Functional images were initially examined for spike artifacts using custom software (<http://www.columbia.edu/cu/psychology/tor/>). Global outlier time points were identified by computing both the mean and the standard deviation of values in each image for each slice. Mahalanobis distances for the matrix of mean values (one per slice) $\times$ functional volumes were computed, and images with a value above 3 standard deviations were considered outliers. Indicators for each individual outlier were entered as regressors in the general linear model (GLM). Next, principal components analysis was used to identify the first 10 components, and component scores were regressed on both outlier-related and task-related regressors (the design matrix; see below). Components that had strong outlier-related correlations and weak task-related (e.g., r^2 values $< .02$) correlations were manually identified and removed from the data.

Preprocessing

After denoising, functional images were slice-acquisition timing and motion corrected using SPM5 software (Wellcome Department of Cognitive Neurology, UCL), using default parameters. Structural T1-weighted images were coregistered to the mean functional image for each subject using an iterative procedure of automated registration using mutual information coregistration in SPM5 and manual adjustment of the automated algorithm's starting point by a trained analyst until the automated procedure provided satisfactory alignment. Structural images were normalized (spatially warped) to a standard template brain (the MNI avg152T1.nii) SPM5's combined segmentation/normalization procedure with default options, and the warping parameters were applied to functional images for each subject. Normalized functional images were interpolated to $2 \times 2 \times 2$ mm voxels and spatially smoothed with an 8-mm Gaussian filter.

First-level GLM model

First-level GLM analyses for each subject were estimated in SPM5. Quantification of single-trial response magnitudes was done by constructing a GLM design matrix with separate regressors for each trial, as in the "beta series" approach of Rissman, Gazzaley and D'Esposito (2004). The model is a three-level mixed effects model, with trial nested within condition at the first level, condition (D vs. ND $\times$ accuracy) within subjects at the second level, and subject as a random effect at

the third level. For each ~ 40 -s trial, we included regressors for (1) the encoding/maintenance period, an 8.5- to 10.5-s epoch, depending on the number of words encoded; (2) the distraction/maintenance period, a 6.8-s epoch; and the probe period, a 6-s epoch. As each trial was modeled individually, separate regression estimates were obtained for each D and ND trial. To assess activity during the encoding/maintenance period *prior* to the onset of distracters, and without assuming a response shape, we modeled epochs as boxcar functions (indicators) without convolving with a hemodynamic response function. All boxcars were delayed for 3 s to compensate for the expected hemodynamic lag. An intercept regressor for each run modeled mean run activity.

The second-level analysis was conducted on first-level regression parameter estimates (activation estimates) for the encoding/maintenance period for each trial. To minimize the effects of MR artifacts on statistical results, trials with images coded as outliers or whose global brain signal values were greater than 3 *SD*s from the mean were excluded (mean = 1.44 outliers per 100 images, *SD* = 0.51). For each voxel within each subject, a generalized least squares model was fit with an AR(1) error structure. Model predictors were distracter type (D vs. ND), accuracy (an integer from 0 to 4), and the distracter type $\times$ accuracy interaction (all predictors were centered). The intercept term in this model corresponded to the overall [encoding/maintenance vs. baseline] contrast, where baseline was the 20-s intertrial interval. These contrast images were subjected to group analysis (the third level of the model). A one-sample *t*-test across subjects (treating subject as a random effect) was performed for each of the four effects separately using robust regression, which increases statistical power and decreases false positive rates in the presence of outliers (Wager, Keller, Lacey & Jonides, 2005).

Search volume and localization of results

Normalized structural T1 images were averaged across participants to create an anatomical underlay for visualizing significant regions of activation and for visually assessing normalization quality. In our experience, this is advantageous because the quality of nonlinear warping can vary across brain regions, resulting in greater differences between the standard brain and subjects' actual T1 images. The average T1 image was segmented using SPM5's unified segmentation/normalization routine, and the union of gray and white matter voxels was used as a mask in all analyses.

Region of interest analyses and multiple comparisons correction

We defined regions of interest (ROIs) on the basis of the conjunction of three criteria: (1) gray/white matter, (2) functional response during the encoding/maintenance period, and

(3) a priori interest based on previous studies. Our a priori hypotheses focused on two sets of regions. One set consisted of regions that were strongly activated in the overall [encoding/maintenance vs. baseline] contrast ($q < .05$, false discovery rate [FDR] corrected, $p < .0095$, see Fig. 2 and Supplementary Table 1) and were additionally located in the lateral or medial PFC, which have been strongly implicated in cognitive control and resistance to distracters. A second set consisted of “default mode” regions (ventromedial PFC, posterior cingulate, superior temporal sulcus [STS]) and nonvisual sensory processing regions (primary somatosensory cortex [S1]/secondary somatosensory cortex [SII], primary auditory cortex [AI]/SII), which are increasingly thought to play a role in task-irrelevant thought (Andrews-Hanna, Reidler, Huang & Buckner, 2010; Mason, Norton, Van Horn, Wegner, Grafton & Macrae, 2007; McKiernan, D'Angelo, Kaufman & Binder, 2006), self-directed attention (Northoff, Heinzl, de Greck, BERPpohl, Dobrowolny & Panksepp,

2006), and poorer cognitive performance (Eichele et al., 2008; Weissman, Roberts, Visscher & Woldorff, 2006) and were also strongly deactivated in the [encoding/maintenance vs. baseline] contrast ($q < .05$, FDR-corrected). Within ROIs, we report results at a threshold of $p < .001$, which is both the modal threshold used in fMRI studies and sufficient to control the family-wise error rate at $p < .05$ across voxels within each ROI. However, for completeness and archival purposes, Supplementary Fig. 1 (see Supplementary Material) shows results for all regions that met criteria (1) and (2) above.

Results

Behavioral

During the MRI session, mean accuracy was 71.21 % ($SE = 1.81$ %) in the ND condition and 61.91 % ($SE = 2.37$ %) in the

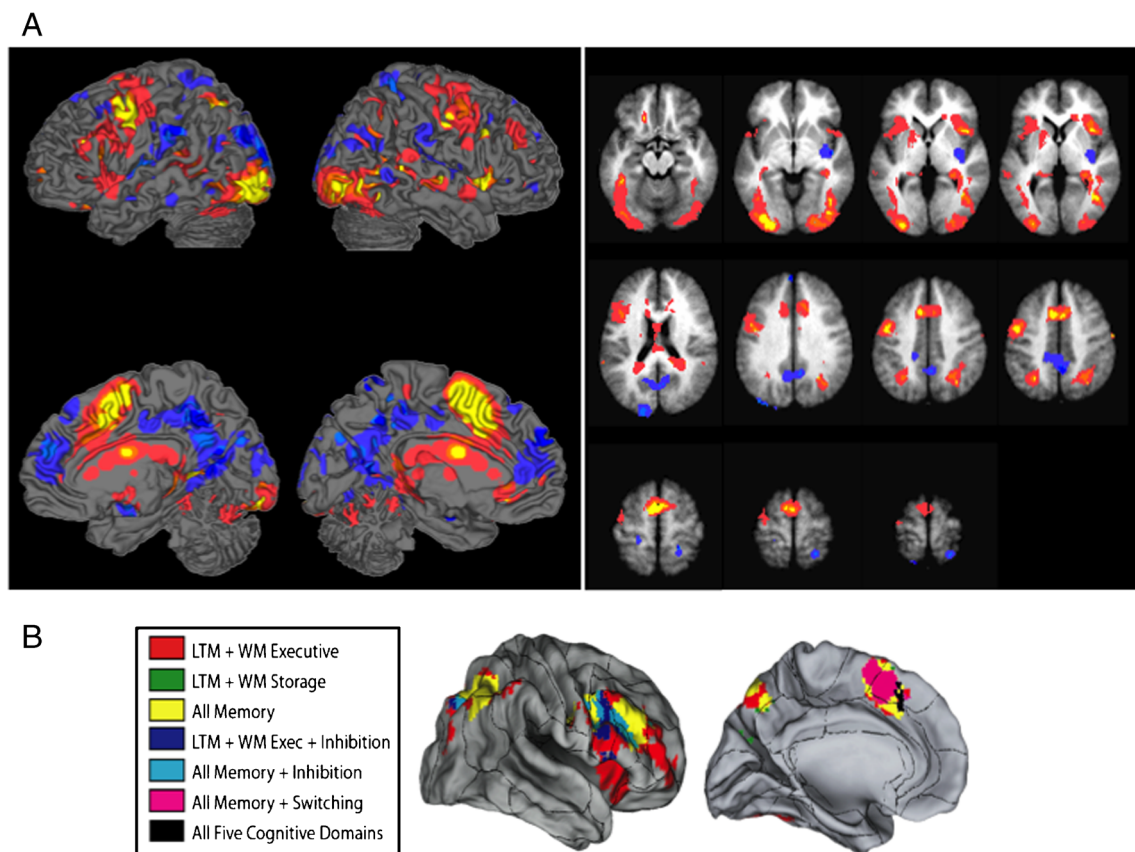


Fig. 2 FMRI activity related to working memory encoding/maintenance. **a** Activations and deactivations in the [encoding/maintenance–baseline] contrast, shown in red/yellow colors and purple/blue, respectively. The primary threshold was $p < 10^{-6}$ (yellow/dark blue for positive/negative effects) and 3 contiguous voxels. Contiguous voxels at lower thresholds that are contiguous with those at the primary threshold are also shown, at $p < 10^{-5}$ (orange/light blue) and $p < .009$ ($q < .01$, false discovery rate corrected, shown in pink/purple). This stringent threshold was chosen to provide anatomically specific regions of interest, although activity in

this contrast was widespread. The anatomical underlay is the average T1-weighted image from this sample, after normalization to Montreal Neurological Institute space. **b** Results from meta-analyses of working memory (WM), long-term memory (LTM), task switching, and inhibition tasks for comparison with panel a. WM is divided into executive (Exec) and storage-only tasks. The black lines show anatomical divisions between Brodmann’s areas as implemented in Caret software (<http://brainvis.wustl.edu/wiki/index.php/Caret>About>). Figure adapted from van Snellenberg and Wager (2009)

D condition. Mean d' was 1.30 ($SE=0.14$) in the ND condition and 0.65 ($SE=0.12$) in the D condition. The results indicate, overall, that participants performed slightly below 75 %, the target level of accuracy determined by the memory set titration task and that they performed slightly better in the ND condition than in the D condition.

Encoding/maintenance versus baseline

Standard WM-related regions were strongly activated during the encoding/maintenance period (prior to distracter presentation; $p < .01$, FDR). As is shown in Fig. 2a (hot colors, yellow/orange/red), activated regions included the bilateral middle and inferior frontal gyri, intraparietal sulcus, inferior temporal cortices, pre-supplementary motor cortices (pre-SMA) and anterior cingulate (ACC). These results correspond closely with those from a recent meta-analysis of WM and related executive tasks (van Snellenberg & Wager, 2009; see Fig. 2b). Deactivation, shown in purple/blue in Fig. 2a, was found in “default mode” regions associated with internal monitoring and task-unrelated thought during rest (Andrews-Hanna et al., 2010; Gusnard & Raichle, 2001), including the anterior medial prefrontal cortex (aMPFC), posterior cingulate (PCC), precuneus, inferior parietal cortex (IPC), and STS. In addition, deactivations were found in areas associated with sensory visual and somatosensory processing, including the primary and secondary visual cortices (VI/VII), extrastriate cortex, dorsal posterior insula (dpINS), and SII. The combined activated and deactivated regions were used as a mask for subsequent analyses of accuracy.

Predictors of subsequent accuracy

We searched within a subset of functionally defined [encoding/maintenance–baseline] ROIs of a priori interest (see the Method section for region definition details) for subsequent accuracy effects. We expected activated areas in the lateral and dorsomedial PFCs to show positive accuracy–performance relationships—that is, greater activity should predict parametric increases in subsequent accuracy—and deactivated “default mode” and somatosensory regions (dpINS and SII) to show negative accuracy–performance relationships.

We found monotonic relationships between brain activity increases and accuracy in three ROIs: the pars opercularis of the inferior frontal gyrus (IFG) bilaterally (BA 44), based on the SPM Anatomy Toolbox v. 1.6 (Eickhoff et al., 2005), and the pre-SMA; both are shown in yellow in Fig. 3. Detailed plots of responses for the IFG and pre-SMA are shown in Figs. 4 and 5, respectively. Further analyses in all the areas that showed significant [encoding/maintenance–baseline] responses revealed that there were additional regions besides these three that were positively correlated with accuracy

(Fig. 3; see Table 1 for statistics), including the premotor cortex, right anterior insula, inferior parietal lobule, left premotor cortex, and left inferior occipital/temporal cortex. These results are consistent with the hypothesis that increases in multiple regions with a distributed network support robust WM encoding and maintenance (cf. Pessoa, Gutierrez, Bandettini & Ungerleider, 2002).

Deactivation in several “default mode” and somatosensory ROIs also predicted subsequent WM accuracy (Table 1; negative relationships, shown in blue in Fig. 3), including the pregenual cingulate/aMPFC, right STS, hippocampus, and bilateral IPC. All of these areas were deactivated during encoding/maintenance overall, and greater deactivation predicted increasing subsequent WM accuracy. In addition, right the SII showed the same pattern of stronger deactivation with increasing accuracy. In the broader search within all [encoding/maintenance–baseline] areas, negative relationships with accuracy were also found in the lateral occipital cortex and several motor/premotor regions. These findings are broadly consistent with the idea that accurate WM encoding and maintenance rely on reduced activation in brain systems that encode task-unrelated perceptual and motor processes (e.g., Andrews-Hanna et al., 2010).

Predictors of distracter-resistant memory

Regions in which activation during encoding/maintenance supports distracter-resistant memory should (1) be activated during encoding/maintenance, (2) *positively* predict accuracy overall, and (3) show a *positive* interaction with distraction, indicating a stronger accuracy effect on D trials relative to ND trials. Thus, to assess regions that differentially predicted accuracy under D and ND conditions, we tested the distraction $\times$ accuracy interaction in the average signal within accuracy-predictive regions discussed above (i.e., regions listed in Table 1).

Only one region, pre-SMA, fit all of the criteria (Fig. 4). Positive accuracy and accuracy $\times$ distracter effects indicated that activity was more positively related to subsequent memory for D than for ND trials ($p < .05$; see Table 1). Furthermore, post hoc tests revealed that the accuracy effect was significant for D trials, $t(16)=2.86$, $p=0.01$, but not for ND trials, $t(16)=-0.33$, $p=.74$, suggesting that pre-SMA activity is important for establishing distracter-resistant memory. In contrast, the bilateral IFG regions that predicted accuracy were equally predictive for D and ND trials, as is shown in Fig. 5 (see Table 1 for statistics).

It is possible that distracter-resistant memories are encoded in regions of the PFC other than our stringently defined a priori ROIs, which required significant [encoding/maintenance–baseline] activity and positive accuracy prediction across all trials. To test this possibility, we searched for voxels that showed accuracy $\times$ distracter effects at a threshold of

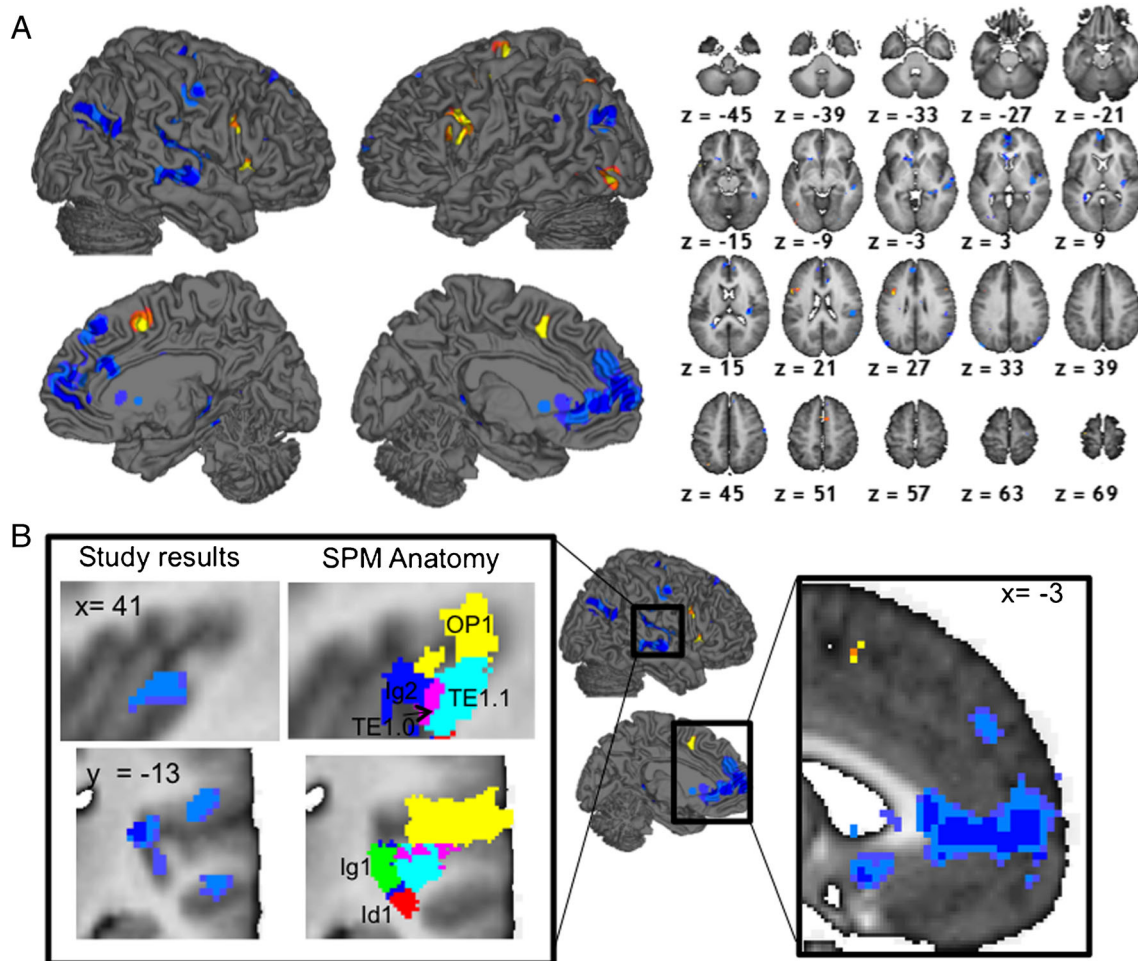


Fig. 3 Regions both responsive to working memory demand and predictive of subsequent accuracy. **a** Left: Surface rendering of areas significant in the conjunction of [encoding/maintenance–baseline] and subsequent accuracy. In yellow/orange areas, activity increases predicted higher accuracy, and in blue/purple areas, activity decreases predicted higher accuracy. Right: axial slices showing results for this conjunction. The threshold is $p < .001$ (yellow/dark blue), $p < .005$ (orange/light blue), and $p < .01$ (pink/purple), with voxels at the latter thresholds contiguous with those at higher thresholds. **b** Left: Detailed localization of sensory

regions, compared with the anatomical regions from the SPM Anatomy Toolbox v.1.7. Deactivations predicting accurate memory were localized in somatosensory areas OP1 (SII) and the dorsal posterior insula (extending into Ig1) and auditory areas TE1.1/1.0. Right: Increases in the pre-SMA and decreases in the ventromedial prefrontal cortex predictive of accurate performance. The full extent of activation is shown for display purposes, although only the more dorsal portion was significant in the [encoding/maintenance–baseline] contrast

$p < .001$ across the frontal cortex. The results revealed both positive and negative accuracy $\times$ distracter interactions. Only one region of the right middle frontal gyrus (MFG; BA 9/46; orange in Fig. 6) showed a positive relationship with performance only in the D condition, as predicted by the distracter-resistance account of lateral prefrontal function. It showed [encoding/maintenance–baseline] activation overall, but at a lower threshold, $t(16) = 3.04$, $p = .008$, and was not predictive of accuracy across all trial types. Other prefrontal regions, including the bilateral MFG (BA 45), premotor cortex (BA 6), and left VLPFC (BA 47/12), showed a negative interaction (blue in Fig. 6). Examining the pattern of these interactions revealed that in the right premotor cortex and VLPFC, the interaction was driven mainly by an association between greater activity and lower levels of accuracy on D

trials, contrary to the distracter-resistance account of lateral PFC function. In the MFG (BA 45), the interaction was driven mainly by stronger positive brain–accuracy relationships on ND trials. These findings are inconsistent with the idea that lateral PFC facilitates distracter-resistant WM.

In addition to activated areas in which the magnitude of increases predicts subsequent memory, we expected that the magnitude of decreases in deactivated areas may predict subsequent memory specifically on D trials. Results for “default mode” ROIs are shown in Fig. 7. Among areas in which deactivation magnitude predicted accuracy across all trials (Fig. 3 and Fig. 7a, left), three regions showed interactions with distracter status (Fig. 7a, right): The right SII and dpINS, associated with somatic sensory perception, and the STS, broadly associated with spontaneous cognition and social

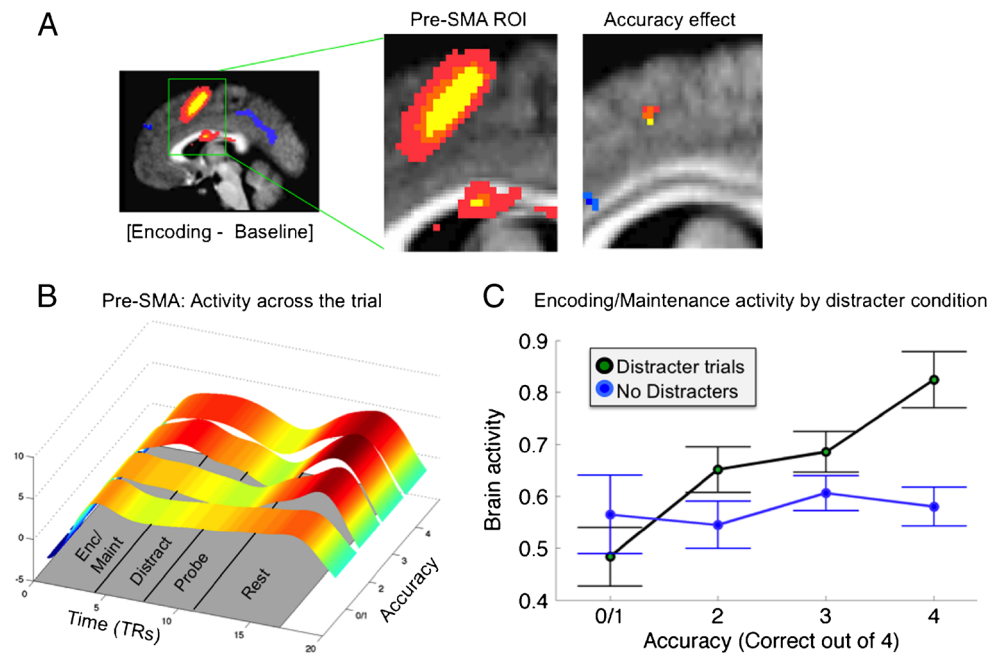


Fig. 4 Presupplementary motor area results. **a** The significant region in used as a functional region of interest (ROI) for distraction analyses. Left: [encoding/maintenance–baseline] results, with thresholds as in Fig. 2. Right: Voxels within this area correlating with accuracy, with thresholds as in Fig. 3, located in the inferior frontal gyrus. **b** Average activity in the accuracy-predictive ROI across the entire trial, for each level of accuracy. During the critical encoding/maintenance period, activity on trials with the highest level of accuracy (three or four correct) is higher than on trials

with lower levels of accuracy. High levels of activation were also found during the probe interval, which is typical for studies of WM and consistent with findings on retrieval-related frontal activation, but is not of primary interest here. **c** Plot of activity in the critical encoding/maintenance period (y-axis) as a function of accuracy (x-axis) and distracter status. The positive overall relationship between brain activity and accuracy is driven by the distracter condition

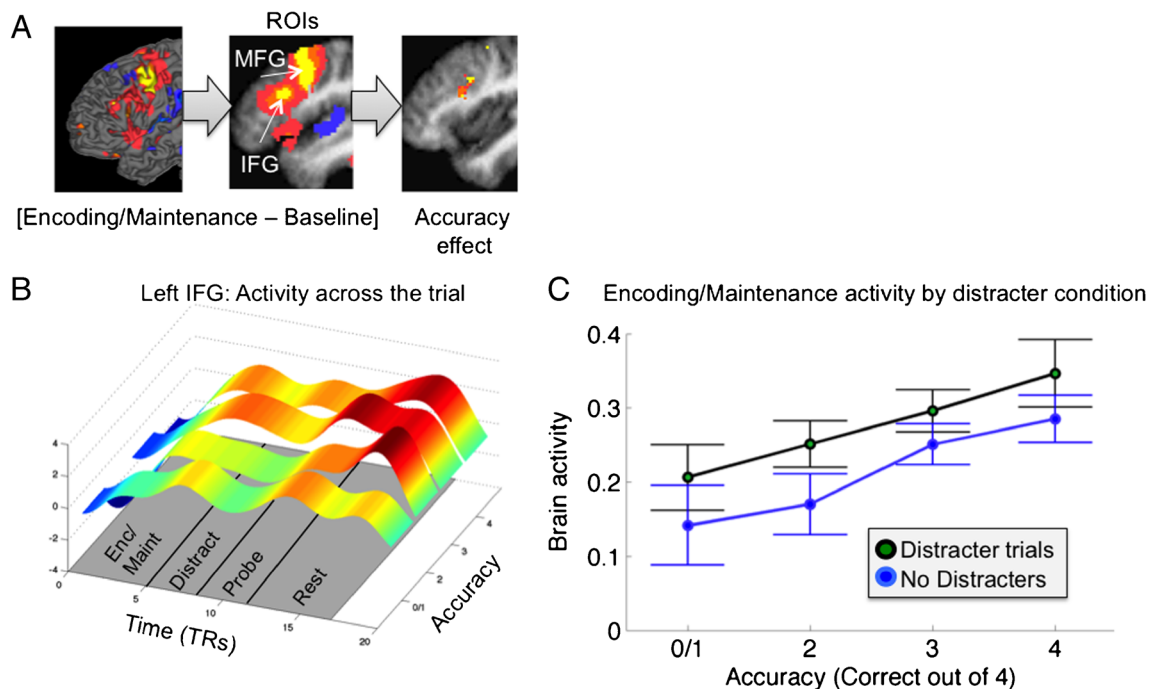


Fig. 5 Ventrolateral prefrontal cortex results. **a** The significant region in used as a functional region of interest (ROI), as in Fig. 4. **b** Average activity in the accuracy-predictive ROI across the entire trial, for each

level of accuracy. **c** Plot of activity as a function of accuracy and distracter status. The relationship between brain activity and accuracy held for both distracter and no-distracter trials

Table 1 Regions activated during encoding/maintenance that predict subsequent accuracy

		Coordinates			Cluster Size		Effect Size	[Encoding/ Maintenance– Base]		Accuracy		Distractor (D/ND)		Accuracy × Distractor	
Region group	Region	<i>x</i>	<i>y</i>	<i>z</i>	Vox	mm ³	Max -log(<i>P</i>)	<i>t</i>	<i>p</i>	<i>t</i>	<i>p</i>	<i>t</i>	<i>p</i>	<i>t</i>	<i>p</i>
Positive: Increasing activation associated with greater accuracy															
Prefrontal regions of interest (ROIs)															
Inferior lateral PFC	IFG/BA44 (L)	−40	10	26	116	928	12.34	6.80	<.0001	2.90	.0052	1.83	.0433	0.00	.4990
	IFG/BA44 (R)	50	14	26	5	40	8.55	3.28	.0024	5.23	.0001	−0.68	.2522	0.79	.2204
Premotor	PMC/BA6 (L)	−40	−4	54	2	16	10.19	3.68	.0010	2.03	.0296	2.61	.0094	−0.57	.2881
	PMC/BA6 (L)	−28	−16	72	5	40	7.48	3.61	.0012	3.41	.0018	1.95	.0347	−0.80	.2166
Medial prefrontal	Pre-SMA	−2	10	50	5	40	6.99	10.48	<.0001	4.44	.0002	1.49	.0778	0.85	.2049
	Pre-SMA (R)	6	8	52	20	160	11.04	7.63	<.0001	3.06	.0038	0.24	.4082	1.86	.0407
Insula/operculum	dAINS (R)	34	20	0	2	16	10.15	6.31	<.0001	4.07	.0005	−0.41	.3425	1.15	.1328
Other regions within [Encoding/Maintenance–Baseline] mask															
Parietal cortex	IPL/PGa (L)	−34	−68	46	7	56	9.57	4.69	.0001	6.55	<.0001	0.47	.3223	0.25	.4034
Temporal cortex	STG (L)	−48	10	14	4	32	8.29	3.89	.0007	3.60	.0012	−0.31	.3792	0.64	.2642
	ITG (L)	−42	−54	−8	16	128	12.12	5.41	.0001	3.52	.0014	−1.56	.0696	1.14	.1358
Occipital cortex	LOCC/V4 (L)	−44	−84	−8	23	184	8.04	6.40	<.0001	3.47	.0016	−0.99	.1681	0.89	.1921
Negative: deactivation or reduced activity associated with greater accuracy															
Default mode/sensory processing ROIs: stronger deactivation associated with greater accuracy															
Ventromedial PFC	pgACC/BA9/ BA32 (R)	10	50	16	17	136	7.17	−3.44	.0017	−3.32	.0022	−1.68	.0564	−0.16	.4361
Rostral DMPFC	DMPFC/BA8 (R)	8	36	48	21	168	7.5	−3.62	.0012	−3.59	.0013	−0.25	.4041	−1.21	.1212
Somatic sensory	SII (R)	50	−26	22	108	864	6.3	−5.23	.0001	−3.64	.0011	−1.67	.0571	4.35	.0003
	DpINS (R)	36	18	8	233	1,864	10.99	−4.86	.0001	−3.76	.0009	−1.05	.1552	2.43	.0137
Inferior parietal	IPL/PGp (R)	52	−66	30	92	736	12.49	−4.39	.0003	−3.52	.0015	0.24	.4079	−0.84	.2055
	IPL/PGp (L)	−48	−76	30	76	608	9.31	−3.56	.0013	−4.62	.0002	−1.15	.1340	−0.42	.3395
	IFL/PF (L)	−66	−46	30	24		9.07	−3.60	.0012	−3.84	.0007	−1.08	.1478	−0.61	.2741
Other regions within [Encoding/Maintenance - Baseline] mask															
Medial temporal lobe	HCMP (R)	30	−36	−2	48	384	7.16	5.73	.0000	−2.99	.0044	−1.75	.0493	0.11	.4562
	HCMP (L)	−34	−28	−6	4	32	8.83	3.73	.0009	−3.82	.0008	−1.11	.1412	−0.24	.4063
	HCMP (L)	−20	−38	10	2	16	8.72	3.68	.0010	−3.23	.0026	−0.02	.4919	−0.11	.4558
Temporal cortex auditory	TOP/TE (R)	52	−6	4	8	64	6.27	−3.39	.0019	−3.00	.0042	−0.56	.2932	1.16	.1323
	ITG (R)	42	−42	−14	63	504	6.9	4.92	.0001	−4.16	.0004	0.57	.2889	−1.27	.1105
Superior temporal	STS (R)	56	−22	−4	132	1,056	7.35	6.98	0.0000	−3.64	.0011	−1.32	.1031	2.33	.0166
Occipital cortex	LOCC/BA17 (L)	−26	−76	6	5	40	8.34	3.54	.0014	−3.96	.0006	−0.50	.3135	0.50	.3104
	LOCC/MTG (L)	−42	−52	8	4	32	7.62	4.79	.0001	−4.47	.0002	0.48	.3194	0.54	.2980
	LOCC/V3/V4 (R)	30	−80	−10	3	24	7.34	3.81	.0008	−3.96	.0006	−1.62	.0626	0.50	.3107
	LOCC (L)	−36	−76	4	4	32	7.4	3.74	.0009	−3.35	.0021	−2.38	.0151	1.62	.0624
Frontal cortex	IFG/BA45/ BA46 (R)	40	28	14	4	32	8.92	4.03	.0005	−4.68	.0001	−0.96	.1751	1.37	.0950
	SMC (R)	60	−12	46	18	144	6.88	5.77	.0000	−3.35	.0021	−0.19	.4265	0.86	.2023
	PMC (R)	30	−16	66	6	48	10.57	3.53	.0014	−5.01	.0001	−0.18	.4302	0.08	.4677
Parietal cortex	IPS (L)	−32	−52	34	1	8	7.7	4.27	.0003	−4.39	.0003	0.07	.4724	0.85	.2027
Medial frontal cortex	MCC (R)	8	−4	26	5	40	6.66	3.64	.0011	−2.42	.0140	−2.09	.0267	1.63	.0612
Anterior cingulate	aMCC (R)	8	28	24	27	216	9.12	3.55	.0014	−5.26	.0001	−1.79	.0460	1.54	.0717
Basal ganglia	Cau (R)	12	26	6	16	128	10.47	4.15	.0004	−8.11	.0000	0.44	.3329	1.05	.1547
	Cau (L)	−12	22	0	189	1,512	9.68	4.20	.0004	−3.39	.0019	2.44	.0133	−0.45	.3291

Accuracy x Distraction Interaction: Frontal Regions

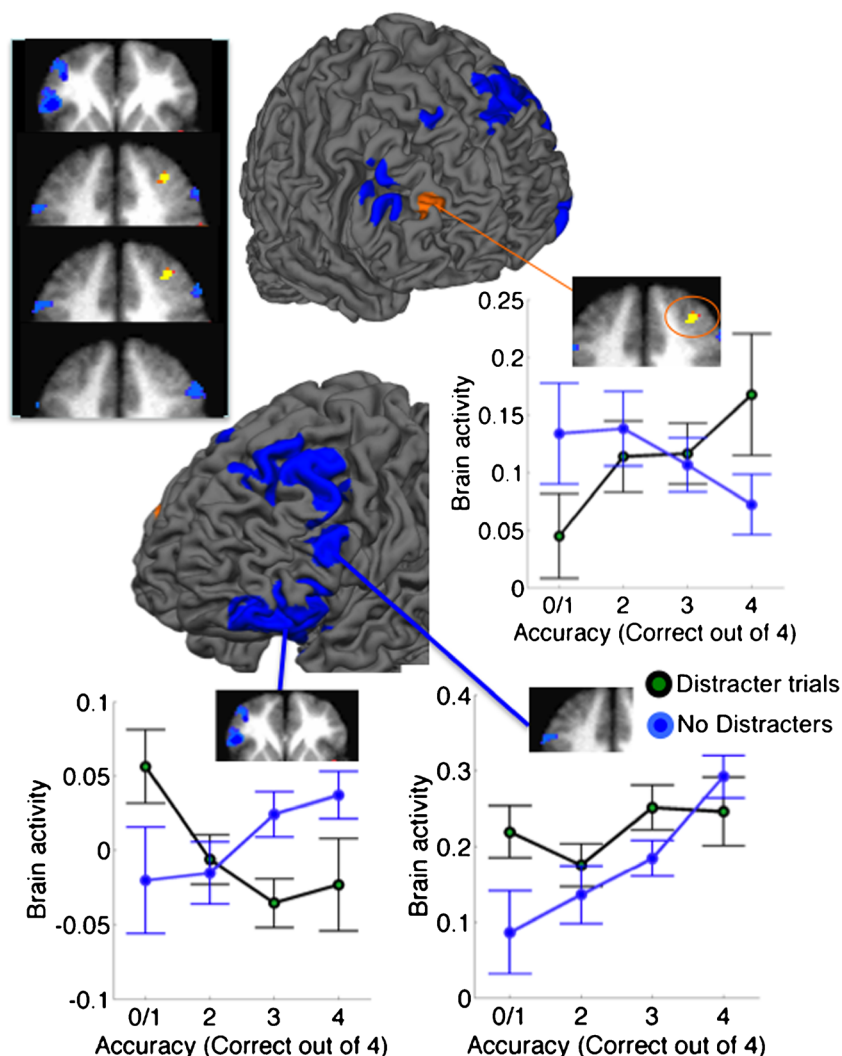


Fig. 6 Accuracy $\times$ distraction interaction results across the functional cortex. These results are not constrained to functional regions of interest. In yellow/orange areas, the activity–performance relationship was more positive for the distracter trials. In blue areas, the relationship was more

positive for the no-distracter trials. Thresholds and colors are as in Fig. 3. Whole-brain results for the accuracy $\times$ distraction interaction are shown in Supplementary Fig. 1

cognition (see the Discussion section). However, the distracter $\times$ accuracy interaction in each of these areas was positive, indicating that decreases in activity were most beneficial for accurate performance in the ND condition. For example, as is shown in Fig. 7b, low accuracy in the ND condition, in particular, was associated with increased SII activity. These results underscore the idea that different processes may be rate-limiting steps in performance with and without distraction.

In contrast, the aMPFC, a part of the “default mode” network, showed a trend toward stronger deactivation with more accurate memory specifically on D trials. Although this effect did not reach significance for the accuracy $\times$ distracter interaction in the ROI defined by overall subsequent accuracy

effects (Table 1), an adjacent region just outside the accuracy-predictive search mask did show an accuracy $\times$ distracter effect at $p < .001$ ($xyz = [6, 58, 8]$, 2 voxels, max $t = 9.50$; shown in Fig. 7c). Thus, deactivation of the aMPFC may also be important for creating distracter-resistant memory.

Discussion

In the present study, we assessed the role of the PFC and other regions in creating memories resistant to dual-task distracters by using a WM paradigm with two novel elements. First, we matched performance on D and ND versions of the task by adaptively titrating the memory set size, ensuring that we

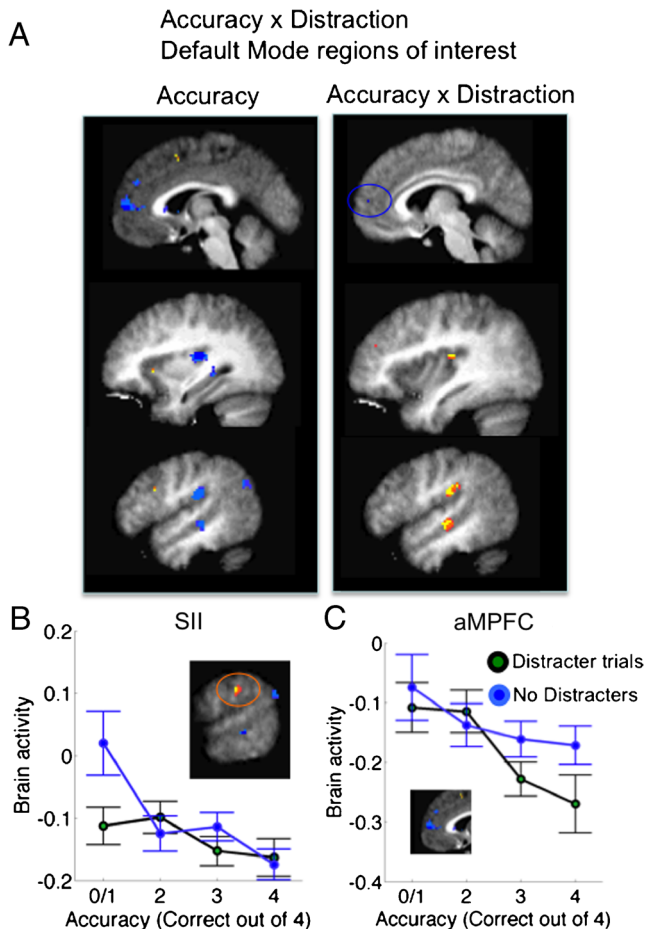


Fig. 7 Accuracy $\times$ distraction interaction results in default mode and sensory regions of interest. **a** Left: Regions predictive of accuracy and significantly activated during encoding/maintenance. Right: Voxels showing accuracy $\times$ distraction interactions. Thresholds and colors are as in Fig. 3. In yellow/orange areas, the activity–performance relationship was more positive for the distracter trials. **b** Detail for secondary somatosensory cortex (SII). In sensory areas, including SII and dorsal-posterior insula, deactivation was more strongly related to performance in the no-distracter condition. **c** In the anterior medial prefrontal cortex (aMPFC), deactivation was more strongly related to performance in the distracter condition

could efficiently estimate brain-activity–performance relationships for both tasks. Second, we used a multipart recognition probe with five levels of accuracy per trial, enhancing our ability to assess activity–performance relationships within subjects. This approach is relatively unique, because in contrast to the hundreds of published WM studies, only a handful have reported relationships between PFC activity and behavioral performance (Gray, Chabris & Braver, 2003; Jha et al., 2004; Pessoa et al., 2002; Rypma, Berger & D’Esposito, 2002; Rypma & D’Esposito, 2003; Smith, Geva, Jonides, Miller, Reuter-Lorenz & Koeppel, 2001).

We replicated earlier results of Sakai et al. (2002) by showing that activation during encoding/maintenance in both the lateral and medial PFC predicted successful subsequent performance under distraction. However, among rigorously defined functional ROIs, only activation in the pre-SMA

predicted performance *specifically* under distraction. Activity in VLPFC predicted subsequent performance in *both* D and ND conditions of the task and was not selective to the D task. An exploratory search outside of the strict boundaries of our functional localizers also revealed an area in the right DLPFC (BA 46) close to results in which activation showed a more positive relationship with subsequent performance under D than under ND conditions. However, activation of nearby and more widespread areas within the bilateral IFG and DLPFC (bilateral BA 45 and left BA 47) predicted accurate performance only in the ND task, in opposition to the distracter-resistance hypothesis of lateral PFC function.

The present results suggest that pre-SMA, which has been activated in numerous WM tasks (Rottschy et al., 2012) but has received less attention in studies of WM than has the lateral PFC, plays a more important role than previously thought in maintaining representations that are resistant to dual-task interference, which is particularly important for complex cognition. Pre-SMA (1) was activated during WM encoding/maintenance, (2) correlated positively with subsequent WM accuracy across all trials on average, and (3) correlated more positively for D than for ND conditions. Furthermore, the activation–performance correlation appeared to be exclusively driven by the D condition. This finding is consistent with a recent meta-analysis of executive function in WM linking distracter resistance to the pre-SMA (Nee et al., 2013) and is interesting because there is substantial evidence that the pre-SMA plays a very general role in goal-directed cognition and action selection across many tasks (Hampshire, Duncan & Owen, 2007; Johnson, Raye, Mitchell, Greene, Cunningham & Sanislow, 2005; Passingham, Chen & Thaler, 1989; Raye, Mitchell, Reeder, Greene & Johnson, 2008; Shima & Tanji, 1998; Stuss & Alexander, 2007; van Snellenberg & Wager, 2009), including preparation (Brass & von Cramon, 2002, 2004; Chikazoe, Jimura, Hirose, Yamashita, Miyashita & Konishi, 2009). Thus, it may be the strength of engagement of basic representation selection and rehearsal processes that create distracter-resistant WM, rather than a circuit dedicated specifically to managing distraction.

Our results also have implications for the role of the lateral PFC in WM and suggest the somewhat surprising conclusion that in contrast to the dominant view (Kane & Engle, 2002; Miller & Cohen, 2001), the lateral PFC may play a limited role in managing dual-task situations in healthy individuals. The fact that multiple areas in the VLPFC and DLPFC, along with temporal regions, predicted performance more strongly in the ND condition implies that memory set representations in prefrontal-temporal circuits are fragile and are disrupted by distracters. Performance correlations identify areas that implement “performance-limiting” processes, or processes that fail intermittently and thus influence accuracy. VLPFC and DLPFC are almost certainly involved in task representation and maintenance, but the critical determinant of accuracy

under dual-task distraction may be another facet of the representation—for example, semantic or multimodal associations that allow the memory to persist. The creation of a more fragile memory representation in the ND condition is also consistent with the idea that activation of task-irrelevant representations (e.g., presumably somatosensory-related activity in S2) might be more disruptive in the ND condition, which was the case here. Whether VLPFC supports an efficient but fragile type of encoding remains to be investigated in future experiments.

This conception of lateral PFC function is not inconsistent with previous work on distraction in WM, provided that the distinction between perceptual distracters and dual-task distraction is acknowledged. For instance, Chao and Knight (Chao & Knight, 1998; Chao, Nielsen-Bohman & Knight, 1995) found that patients with lateral PFC damage showed impaired WM performance with perceptual distracters (auditory tones) and increased electrophysiological responses to such tones, suggesting reduced distracter inhibition. In monkeys, Miller et al. (1996) reported that activity in prefrontal neurons persisted across a distracter-filled delay period, but activity in inferior temporal neurons was disrupted by distracter presentation. In humans, Jha et al. (2004) reported greater lateral PFC activity during WM maintenance with concurrent presentation of perceptual distracters that were similar to the memory items. These findings and others support the idea that LPFC is important for stimulus selection and reduction of proactive interference (Badre & Wagner, 2005; Ferredoes, Tononi & Postle, 2006; Fuster, 2008; Nee & Jonides, 2009; Nelson, Reuter-Lorenz, Sylvester, Jonides & Smith, 2003; Thompson-Schill, D'Esposito, Aguirre & Farah, 1997)—that is, keeping irrelevant representations out of mind—but not management of concurrent task sets required for resistance to dual-task distraction. Since very few studies have examined the neural bases of dual-task distraction (for a discussion, see Chein, Moore & Conway, 2011) and management of dual-task distraction is particularly predictive of complex cognitive abilities (Kane, Hambrick, Tuholski, Wilhelm, Payne & Engle, 2004; Miyake et al., 2000), the present results serve as a launching point for explorations of the brain systems most critical for enabling complex cognition.

Here, another set of findings that argue against a lateral-PFC-dominated view of WM maintenance is the robust relationship between WM performance and deactivation of "default mode" and sensory processing regions. "Default mode" regions are associated with mentalization about the self (Northoff et al., 2006), autobiographical memory retrieval (McDermott, Szpunar & Christ, 2009), affective arousal (Wager et al., 2008), emotion (Wager et al., 2008), and task-irrelevant thoughts (Andrews-Hanna et al., 2010; Christoff, Gordon, Smallwood, Smith & Schooler, 2009; Mason et al., 2007; Wang, Yu, Xu, Qin, Li & Jiang, 2009). These findings parallel similar relationships between "default mode"

decreases and performance on basic attention tasks (Eichele et al., 2008; Weissman et al., 2006) and long-term memory encoding (Daselaar, Prince, Dennis, Hayes, Kim & Cabeza, 2009).

Thus, although lateral PFC increases may be a robust feature of WM processes and an intact PFC may be critical for complex memory operations, the lateral PFC may be "necessary but not sufficient" for good performance in the context of dual-task distraction. In contrast, activation in pre-SMA (likely related to response generation) and deactivation of regions involved in spontaneous, internally generated cognition may be critical.

Limitations and methodological considerations

This study was designed to assess brain–performance relationships for distracting versus nondistracting WM conditions and, so, required that performance be matched in the two conditions and that the task be difficult enough to be in a sensitive range for brain–performance correlations for each subject. To do this, we opted to titrate the memory set size for each subject for each task, resulting in a higher memory set size for the ND than for the D condition. Because of this, we did not attempt to compare the main effects of the D versus ND conditions on brain activity. Previous studies have supplied this information, reporting increased activity in the DLPFC (Jha et al., 2004; Postle, 2005), MFG (McNab & Klingberg, 2008), and VLPFC (Jha et al., 2004)—for example, during processing of task-relevant distracters.

Additionally, we did not opt to titrate the number of distracters. As is stated above, we varied the size of the memory set in order to match performance on D and ND trials and provide a relatively high-powered assessment of D versus ND $\times$ accuracy effects. We did not titrate the number of distracters, since doing so would have introduced variance across individuals in the WM delay period, which we believed would introduce additional unwanted complexity into the design, since the delay duration substantially impacts WM performance. Thus, we cannot ensure that the distracters were equally distracting for all participants, but this did not affect our ability to conduct the within-subjects contrasts (accuracy, D vs. ND $\times$ accuracy) of primary interest. It might have been possible to design a titrated-distracter version of the task, but this was not mandated by our hypotheses.

Another design choice was to block the D and ND conditions, thus preventing confusion and avoiding task switching on the part of participants. Because participants knew whether the distracters would appear or not, they could easily adopt different strategies (based possibly on differing expectancies and/or preparation processes) in D versus ND conditions. Thus, we cannot tell whether it is the type of representation or the type of strategy employed that causes different regions to predict performance in each case. However, we regard this

as an important and ecologically valid feature of the task: Knowing that one will be distracted affords the opportunity to engage in robust encoding and maintenance processes, strategic or otherwise. Such knowledge is a feature of complex WM span tasks that predict general fluid intelligence. Future studies could specifically investigate the impact of knowledge about upcoming distraction.

Fourth, it might be argued that the high WM load in our task produced "distraction" in the D and ND conditions alike. However, the double dissociations in the brain–performance correlations in D and ND conditions suggest that these two conditions are not equally distracting, or at least not in the same ways. As we argued above, interference due to memory load should be thought of as conceptually distinct from the requirement to manage a demanding secondary task.

Fifth, it might be argued that ours was not a WM task, given that we presented up to 11 two-syllable words, which may go beyond what is considered WM capacity, and given that we used probes to test memory at the end of each trial, raising the possibility that high performance was achieved by means of long-term memory or non-WM strategies. In light of these potential concerns, we note the following ideas: (1) The task used a standard delay and probe for WM based on Sternberg's classic work; (2) the task elicited activation in standard WM brain areas; (3) potential chunking of verbal material in WM is consistent with capacity limits; and (4) the task was not designed to excise long-term memory processes—indeed, long-term memory processes have been implicated in WM in many situations. Thus, we believe this task was one that tested WM.

Sixth, we did not attempt to separate encoding and maintenance processes, because doing so would have come at a substantial cost in power for our main fMRI comparisons. Future studies might fruitfully separate these processes.

Finally, the task we used was designed to study the mechanisms of WM maintenance but is not isomorphic with the tasks and task parameters shown to predict fluid intelligence. Studying such tasks with fMRI (Chein et al., 2011) and correlating fMRI activity with fluid intelligence directly (Gray et al., 2003) is an important future direction.

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